

Face recognition device browser UI interface document

- Based on HTTP Protocol Communication
- Document Revision 2.0
- Release Date December 9th, 2022
- Modifier Lai Jinjie

Updated Records

Version No.	Modify the date	*Description

HTTP Protocol Description

- **Support HTTP 1.1
- GZIP Compression Techniques

API Unified interface return format

Returns format field description

Field	Type	Length	Required	Description
result	bool	5	Yes	Return result status true/false
content	object	1	Yes	Return result load
errCode	int	4	Yes	Return the error code when the operation is wrong
error	string		No	Error description text

Return format example

```
{
  "result": false,
  "content": null,
  "errCode": 401,
  "error": "Unauthorized"
}
```

Heartbeat

Brief Description:

- Heartbeat interface, used to check whether the network connection is normal.

Request URL:

- Example: <http://192.168.1.150/api/heartBeat>
- <http://Device> IP:port/api/heartBeat
- Example: <http://192.168.1.150/api/heartBeat>

Request Mode

- GET

Return value description

Content-Type

- application/json; charset=utf-8

```
{
  "result": true,
  "content": "OK"
}
```

Obtain Device SN

Brief Description:

- Heartbeat interface, used to check whether the device network connection is normal

Request URL:

- <http://Device> IP:port/api/GetDeviceSN
- Example: <http://192.168.1.150/api/GetDeviceSN>

Request Mode:

- GET

Return value description

Content-Type

- application/json; charset=utf-8

```
{
  "result": true,
  "content": "Device SN"
}
```

User Management

API-Login

Breif Description

- When you first open the device browser page, call this interface to generate a Login Token for subsequent access to the API interface.
- The Token after login will remain valid for 24 hours
- If you enter the wrong password 5 times during login, the system will prohibit you from logging in within 5 minutes
- The device can store up to 100 tokens

Request URL:

- <http://Device IP:port/api/User/Login>
- Example: <http://192.168.1.150/api/User/Login>

Request Mode:

- POST

Content-Type

- application/json

Request parameters

Field	Type	Length	Required	Description
password	string	32	Yes	Device login password It is also the menu admin password Password not transmitted in plaintext Password is MD5 calculated by adding salt to a random string
Hash	string	n	Yes	Add salt to password

Request parameter example

```
{
  "password": "31BF648EE6FBFC1B807E485089AFC692",
  "Hash": "95e17dcb-4221-494d-8586-6d5e8b077761"
}
```

```
//Adding salt to passwords to confuse the calculation of hash values
std::string strAdminPassowrd; //Actual password
std::string sHash; //Random string
std::string sHashPassword; //Password string after adding salt
sHash = CreateGUIDString();//Create a random string *****to obtain the
value of the Hash field*****

//Start adding salt to confuse
sHashPassword.append(sHash);
sHashPassword.append(strAdminPassowrd);
sHashPassword.append(sHash);

// Use sHashPassword to calculate the MD5 value of a string and encode the MD5
value into a HEX string
```

```
// *****Obtain the value of the password field*****
sHashPassword = CreateMD5HexString(sHashPassword);

### **Return value description**

Returns the JWT Token for subsequent access

**Return Value Example**

```json
//Returned when verification succeeds
{
 "result": true,
 "content": {
 "token": "eyJhbGciOiJodHRwOi8vd3", //token character string

 "expiration":123456789 //token expiration time
 }
}
```
```

```
```json
//Return if validation fails
{
 "result": false,
 "errCode": 1,
 "error": "Wrong Password!"
}
```
```

****Wrong Code****

| Code | Description |
|-------|--|
| :--- | :----- |
| 1 | Wrong Password! |
| 2 | Password error too many times, locked! |
| 10001 | Wrong method requested |
| 10002 | Request Content-Type error |
| 10003 | Request body, gzip decompression error |
| 10004 | The Json format in the request body is incorrect |

API-log out

****Brief Description:****

- When you need to log out of the system, calling the logout function will delete the corresponding Token.

****Request URL:****

- `http://Device IP:port/api/User/Logout`
- Example: `http://192.168.1.150/api/User/Logout`

****Request Mode:****

- Get

****Request parameters**

- None

****Return value description****

****Return value example****

```
```json
//Returned when verification succeeds
{
 "result": true,
 "content": "ok"
}
```
```

API-Verify whether the Token is invalid

****Brief Description:****

- API for checking if the token has expired after login

****Request URL: ****

- `http://Device IP:port/api/User/CheckLoginToken`
- Example: `http://192.168.1.150/api/User/CheckLoginToken`

****Request Mode:****

- GET

****Verify Token****

- Verify

****Content-Type****

- application/json

****Authorization****

- Bearer token character string

Return value description

Return Token expiration time, unix timestamp

****Return value example****

```json

//Returned when verification succeeds

```
{
 "result": true,
 "content": 1234567899
}
```

```

```json

//Returned when verification fails

```
{
 "result": false,
 "errorCode": 10000,
 "error": "Token is invalid"
}
```

```

****Wrong Code****

Code	Description
10000	token is expired, please log in again
10001	wrong request method

--	--

API-Renew for Token

****Brief Description:****

- Renew the Token that is about to expire

****Request URL:****

- http://Device IP:port/api/User/TokenExtension
- Example: http://192.168.1.150/api/User/TokenExtension

****Request Mode:****

- GET

****Verify Toekn****

- verify

****Content-Type****

- application/json

****Authorization****

- Bearer token character string

****Return value description****

Return Token expiration time, unix timestamp

******Return value example******

```
```json
//Returned when verification succeeds
{
 "result": true,
 "content":
}
```
```

```
```json
//Returned when verification fails
{
 "result": false,
 "errCode": 10000,
 "error": "Token is invalid"
}
```
```

****Wrong Code****

| Code | Description |
|-------|---------------------------------------|
| 10000 | token is expired, please log in again |
| 10001 | wrong request method |

API-Modify the management password

****Brief Description:****

- Changing the device's management password will also change the menu password

****Request URL:****

- http://Device IP:port/api/User/EditPassword

- Example: `http://192.168.1.150/api/User/EditPassword`

****Request Mode:****

- POST

****Verify Toekn****

- Verify

****Content-Type****

- application/json

****Authorization****

- Bearer token character string

Request parameters

| Field | Type | Length | Required | Description |
|-------------|------|--------|----------|--------------------|
| :----- | :--- | :--- | :--- | :----- |
| OldPassword | int | 8 | Yes | Old login password |
| NewPassword | int | 8 | Yes | New login password |

****Request parameter example**

```
```json
{
 "OldPassword": "123456",
 "NewPassword": "888888"
}
```
```

Return value description

****Return value example****

```
```json
//Returned when verification succeeds

{
 "result": true,
}
```
```



```

```json
//Returned when verification fails
{
 "result": false,
 "errCode": 1,
 "error": "old password error"
}
```

```

****Wrong Code****

| Code | Description |
|-------|--|
| :--- | :----- |
| 10000 | token is expired, please log in again |
| 10001 | Wrong request method |
| 10002 | Request Content-Type error |
| 10003 | Request body, gzip decompression error |
| 10004 | The Json format in the request body is incorrect |
| 1 | Old password incorrect |
| 2 | The new password format is incorrect. It must be a number
and cannot exceed 8 digits, cannot be less than 4 digits, and cannot be empty |

API-Check if the device's management password is empty

****Brief description: ****

- Check if the device's management password is empty. If it is empty, please call /api/User/IniAdminPassword to initialize the management password
- The device was not configured with a management password at the factory

****Request URL: ****

- http://Device IP:port/api/User/CheckAdminPasswordIsEmpty
- Example: http://192.168.1.150/api/User/CheckAdminPasswordIsEmpty

****Request method: ****

- GET

****Verify Toekn****

- Not verifying

****Content-Type****

- application/json

Return value description

Return value example

```
```json
{
 "result": true,
 "content": {
 "IsEmpty": 1 //1-- indicates that the password is empty. 0-- indicates
that a password has been configured
 }
}
```
```

API-Initialize management password

**Brief description: **

- When the management password of the device is empty, initializing the management password of the device will also modify the menu password
- Can only be called when the device management password is empty

**Request URL: **

- `http://Device IP:port/api/User/IniAdminPassword`
- Example: `http://192.168.1.150/api/User/IniAdminPassword`

**Request method: **

- POST

Verify Toekn

- Not verifying

Content-Type

- `application/json`

Request parameters

| Field | Type | Length | Required | Description |
|----------|------|--------|----------|--------------------|
| :----- | :--- | :--- | :--- | :----- |
| Password | int | 32 | Yes | New login password |

****Example of Request Parameters****

```
```json
{
 "Password":"123456"
}
```
```

**Return value description**

**Return parameters**

| Field | Type | Length | Required | Description |
|------------|--------|--------|----------|--|
| : | : | : | : | : |
| token | string | 32 | Yes | After successfully initializing the password, will automatically log in and return the login token |
| expiration | int | | Yes | Indicates the validity period of the token. unix timestamp |

****Return value example****

```
```json
//Return upon successful initialization
{
 "result": true,
 "content": {
 "token": "abcd",
 "expiration": 123456,
 }
}
```
```

```
```json
//Return when verification fails
{
 "result": false,
 "errCode": 1,
 "error": "old password error"
}
```
```

****Error code****

| Code | Description |
|-------|---|
| : | : |
| 10001 | The requested method is incorrect |
| 10002 | Request Content Type error |
| 10003 | Request body, gzip decompression error |
| 10004 | JSON format error in the requested body |
| 1 | Password field format error |

- | 2 | Password field format length error |
- | 3 | The configured management password cannot be initialized |

Device Management

Device Working Parameters

Basic Info of Device SystemInfo

| Field | Type | Description |
|--------------------|--------|--|
| :----- | :--- | :----- |
| DeviceSN | string | Serial No. |
| DeviceName | string | Device Name |
| | | |
| FirmwareVersion | string | Firmware Version |
| FingerprintVersion | string | Fingerprint Algorithm Version |
| FaceVersion | string | Face Algorithm Version |
| | | |
| Manufacturer | string | Manufacturer |
| | | |
| ManufacturerPhone | string | Manufacturer Telephone |
| Website | string | Website |
| ProductionDate | string | Production Date |
| OEMText | string | OEM custom text, up to 200 characters can be filled in |
| | | |
| AutoRestart | int | Daily automatic restart switch |
| AutoRestartTime | string | Automatic restart time every day, format HH:mm |

Device Working Status

| Field | Type | Description |
|---------------|------|--------------------------------|
| :----- | :--- | :----- |
| RunDays | int | Days of system operation |
| FormatCount | int | Formatting times |
| WatchDogCount | int | Number of watchdog reset times |

```

| BootTime | int | Boot Time Unix timestamp (seconds) |
| RelayStatus | int | Relay status<br>0--indicates COM and NC are normally
closed<br>1--indicates COM and NO are normally closed |
| KeepOpenStatus | int | Normally open state<br>0--indicates normally
closed<br>1--indicates normally open |
| DoorSensorStatus | int | Door sensor status<br>0--indicates closed<br>1--
indicates open |
| LockDoorStatus | int | Lock status<br>0--indicates unlocked<br>1--indicates
locked |
| AlarmStatus | string | Door alarm status<br>Empty string means no alarm,
otherwise there will be a specific alarm name<br>fire--fire
alarm<br>blacklist--blacklist alarm<br>anti--anti-dismantling
alarm<br>illegal--illegal authentication<br>password--duress alarm
password<br>openTimeout--door opening timeout alarm<br>doorSensor--door
magnetic alarm<br>when there are multiple alarms, use commas to separate
fire,blacklist |

```

Region and language Language

```

Field	Type	Description
Language	int	Language 1 - Chinese;  2 - English; 3 - Traditional
Chinese;4 - French;5 - Russian<br>6 - Portuguese;7 - Spanish;8 - Italian;9 -
Japanese<br>10 - Korean;11 - Thai ;12 - Arabic;13 - Portugal<br>14 - Turkish ,15
- Indonesian,16 - Ukrainian,17-Vietnamese |
| SystemTime | int | Device time Unix timestamp (seconds) |
| UseNTP | int | Enable NTP automatic time calibration 1--enable; 0--disable |
| TimeZone | int | Device time zone, value range:-12 - +14 |
| Volume | int | Volume (range 0-10) |
| Voice | int | Voice playback switch 0, no playback; 1, playback |

```

Human-machine interaction UI

```

Field	Type	Description
DisplayBrightness	int	Screen brightness Settings 1-10
MenuPassword	string	Menu password, pure numbers, 4-8 digits or blank
ShowIR	int	Display infrared images on the device 1-- Enable; 0-- Disable
ShowPersonPhoto	int	Display the person's portrait after recognition 1--
enable; 0--disable		
PlayPersonName	int	Announce the name of the person after recognition 1--
enable; 0--disable		
RecognitionButon	int	Click the recognition button before recognition 1--
Enable; 0--Disable		
Unregisteredwarn	int	Unregistered personnel reminder 1--Enable; 0--Disable
ShowPersonName	int	whether to display the person's name after
identification 1--Enable; 0--Disable		
FillLight	int	Fill light mode: 0: Normally closed; 1: Normally open; 2:
Automatic;		
UseQRCode	int	QR code recognition switch 1--Enable; 0--Disable
UseFastRecognition	int	Quick recognition, without displaying recognized
names and photos <br /> 1--Enable; 0--Disable |

```

| UseRequestAuthorization | int | Online authentication
 1--Enable; 0--Diabile
After activation, the door will not open after recognition verification
The device needs to request the server to verify again |

Data Storage Storage

| Field | Type | Description |
|-------------------|-------|---|
| RecordAutoCycle | int | Record full cycle 1--record full cycle, 0--record full no cycle, waiting to be cleared |
| SaveUnregistered | int | Save unregistered personnel records, 0: do not store, 1: store
Unregistered personnel refers to personnel who are not registered in the system during face recognition,
or card numbers that are not registered in the system when swiping a card |
| SaveRecordPicture | int | Save the scene picture 0, do not save; 1, save |
| PeopleStorageInfo | class | Personnel storage details |
| RecordStorageInfo | class | Record storage details |

- **Personnel Storage Info peopleInfo **

```json

```
{
{
"Person": {"Max": 5000, "Current":0 },//Personnel storage capacity Max maximum capacity; Current current storage quantity
"Face": {"Max": 5000, "Current":0 },//Face Storage Capacity
"Card": {"Max": 5000, "Current":0 },//Card Capacity Storage
"Fingerprint": {"Max": 5000, "Current":0 },//Fingerprint storage capacity
"PalmVein": {"Max": 5000, "Current":0 },//Palm print storage capacity
"Pasword": {"Max": 5000, "Current":0 },//Password storage capacity
"Admin": {"Max": 5000, "Current":0 }//Administrator storage capacity
}
```
```

- **Record storage information recordInfo**

```json

```
{
"VerifyRecord": {"Max": 5000, "Current":0 },//Access record storage capacity
"DoorRecord": {"Max": 5000, "Current":0 },//Door sensor record storage info
"SystemRecord": {"Max": 5000, "Current":0 },//System record storage capacity
"RecordPhoto": {"Max": 5000, "Current":0 }//On-site photo storage capacity
}
```
```

Face Recognition

Field	Type	Description
-------	------	-------------

Field	Type	Description
FaceIR	int	Liveness detection, 1 on, 0 off
FaceIRThreshold	int	Liveness detection threshold 1-99
FaceDistance	int	Recognition Distance 1--Short Distance(0.2-0.5m); 2--Middle Distance(0.2-1.5m);3--Long Distance(0.2-1.5m or above)
FaceThreshold	int	Face recognition threshold 1-99 The larger the face recognition threshold, the higher the accuracy
FPComparison	int	Fingerprint comparison threshold value range:1-100
FaceMask	int	Mask Detection
FaceMaskThreshold	int	Mask threshold 1-99, the larger the face recognition threshold, the higher the accuracy

Body Temperature Detection

Field	Type	Description
UseBodyTemperature	Int	Temperature measurement mode switch. 0: Non-temperature measurement mode 1: Temperature measurement mode
UseFahrenheitDisplay	int	Turn on Fahrenheit temperature display, 1: on, 0: off
TemperatureCompensate	float	Temperature compensation value -10.0 -- +10.0
TemperatureAlarmThreshold	float	Temperature alarm threshold example 37.5
TemperatureDisplay	int	whether to display body temperature `` 0-disable; 1-enable

Server Parameter NetworkServer

Field	Type	Description
UseTCPClient	Int	Use TCPClient to connect to the server 1--enable; 0--disable;
UseUDPClient	Int	Use UDPClient to connect to the server 1--enable; 0--disable;
ServerAddress	string	Server address TCP or UDP protocol server address
ServerPort	int	Server port number
KeepaliveTime seconds	int	keep-alive packet interval 1-65535

UseHTTPClient	int	whether to enable HTTPClient protocol
1--enable; 0--disable;		
HTTPClient_ServerAddr	string	HTTP protocol server address
HTTPClient_KeepaliveTime	int	HTTP protocol keep-alive packet interval
HTTPClient_UseGZIP	int	whether to use GZIP compression when requesting HTTP protocol
0--do not use; 1--use		
HTTPClient_ProtocolType	int	HTTPClient Protocol Type
100 --- HTTPv1 200 --- HTTPv2		
UseMQTTClient	int	whether to start the MQTTClient protocol
1--enable; 0--disable;		
UseMQTTSSL	int	whether to enable SSL secure socket for MQTT
1--enable; 0--disable;		
MQTTServerAddr	string	MQTT Server Address www.abc.com
MQTTPort	int	MQTT Server port number
MQTTLoginName	string	User name for logging in to the MQTT protocol
MQTTLoginPassword	string	Password for logging in to MQTT protocol
MQTTPublishTopic	string	The topic used by devices to send data in the MQTT protocol
MQTTSubscribeTopic	string	Topic In the MQTT protocol, the topic that the device needs to subscribe to in order to receive data
MQTT_KeepaliveTime	int	The keep-alive packet interval of the MQTT protocol
MQTT_UseGZIP	int	whether to use GZIP compression for mqtt
0--do not use; 1--use		
UsewebsocketClient	int	whether to start websocketClient protocol
1--enable; 0--disable;		
websocketClient_ServerAddr	string	websocket protocol server address ws://192.168.1.1/websocket or wss://192.168.1.1/websocket
websocketClient_KeepaliveTime	int	websocketClient protocol keepalive packet interval
websocketClient_UseGZIP	int	websocket whether to use GZIP compression
0--do not use; 1--use		
websocketClient_ProtocolType	int	websocket protocol type
UseYZW	int	whether to enable Yunzhu network HTTPClient protocol
1-- Enable; 0-- Disable;		
YZWAddr	string	Yunzhu platform protocol server address
YZW_NotUploadRecord	int	Yunzhu Platform Protocol Prohibit uploading records
1--Prohibit; 0--Allow;		
YZW_NotUploadUserPhoto	int	Yunzhu Platform Protocol Prohibit uploading recent photos of workers
1--Prohibit; 0--Allow;		

ClientCert	string	Client certificate PEM format
ClientCertKey	string	Client certificate key PEM format
CACert	string	The CA certificate of the server is used to verify the legitimacy of the server in PEM format. After setting the server CA, the client certificate must also be issued by the same CA

###

Device Network Parameters Network

Field	Type	Description
ConnectPassword	string	Device communication password 32 characters
UseWired	int	wired network switch, 1: on, 0: off
WiredDHCP	int	wired network automatic ip, 1: on, 0: off
WiredIP	string	wired network IP address ("//192.168.0.110")
WiredIPMask	string	wired network subnet mask ("//255.255.255.0")
WiredGteway	string	wired network gateway ("//192.168.0.1")
WiredDNS	string	Dns("//192.168.0.1")
WiredMAC	string	wired network MAC address
UseWIFI	int	wireless network switch, 1: on, 0: off
WIFIAPName	string	wireless network account
WIFIAPPassword	string	wireless network password
WIFIMAC	string	wireless network MAC address
WIFIDHCP	int	wireless network automatic ip, 1: on, 0: off
WIFIIP	string	wireless network IP address("//192.168.0.110")
WIFIIPMask	string	wireless network subnet mask("//255.255.255.0")
WIFIGteway	string	wireless network gateway ("//192.168.0.1")
WIFIDNS	string	wireless network Dns ("//192.168.0.1")

UseWebPage	int	Web page management switch, 1: on, 0: off
HTTPPort	int	Web management page port number, 1-65534
HTTPSPort	int	Web management page port number, 1-65534
WebPageUseSSL	int	Enable SSL on the device web page. Use openssl self-signed SSL certificate
UseUDP	int	UDP port switch, 1: on, 0: off
UDPPort	int	UDP port number (used by UDP protocol)
UseTelnet	int	Linux Telnet function switch, 1: on, 0: off
TelnetPort	int	Telnet Port Number
UseSSH		Enable Linux SSH 1:On,0:off
SSHPort		SSH port number
SSHLoginPassword		SSH or Telnet login password
UserTSP	int	RTSP video stream, 1: on, 0: off
RTSPPort	int	RTSP Port Number
RTSPUser	string	RTSP User Name
RTSPPassword	string	RTSP Password

Access Control Parameters Door

Field	Type	Required	Description
CardBytes	int	Yes	Card number byte; 3, 4, 8; 0--disable card reading
AccessType	int	Yes	Entry and exit type 0, entry; 1, exit
WgFormat	int	Yes	WG output format 26 / 34 / 66
WGContent	int	Yes	WG output content: 1--User ID, 2--Card Number
ReleaseTime	int	Yes	Door open hold time 0-65535 (s). 0 means 0.5 seconds
DelayOpenDoorTime	int	Yes	Delay unlocking time 0-65535 (s). 0--disable
FreeOpen	int	Yes	Open door without verification 1--Enable; 0--Disable
OpenInterval	int	Yes	Repeat recognition interval 0--disabled; 1-65535 (ms)

OpenInterval_SaveRecord	int	Yes	Repeat interval record storage setting 0, do not save; 1, save
Relay	int	Yes	whether the relay supports bistable state. 1 -- support, 0 -- not support
ShortMessage	string	Yes	Short message after legal verification
VerificationType	int	Yes	Verification method 1. Standard mode; 2. **Face***/Fingerprint/Palm vein/Card** **+** **Password** For details, please see the description below
OverdueRemind	int	Yes	Access level expiration reminder 1--enable; 0--disable
OverdueRemind_Day	int	Yes	Access level expiration reminder validity threshold, the minimum remaining valid days. If the number is lower than this, a prompt will be displayed after recognition that the validity period is about to expire. The value range is: 1-255. 0--means closed.
TimingOpen	int	Yes	Timed normally open function 1--Enable; 0--Disable
TimingOpen_mode	int	Yes	Timed normally open. Automatic opening mode: ``1. After legal authentication, it can be normally open within the specified time period``2. If the authorization is marked as normally open, it can be normally open after authentication within the specified time period``3. Automatic opening and closing, automatically opening and closing the door at the time
TimingOpen_timegroup	object	Yes	Timed normally open. Normally Open Period Use Weekly Period Structure
TimingLocked	int	Yes	Timing lock function 1--Enable; 0--Disable
TimingLocked_timegroup	object	Yes	Timed lock. Lock period Use weekly period structure
VisitorRootPassword	string	Yes	Visitor Root Password
MultiPerson	int	Yes	Multi-group door opening ,number of people ;1-50;
DailyLimit	int		Daily frequency limit, 0--disable this function; 1-- Enable this function
DelayOpenDoorTime	int		Delay unlocking time 0-65535 (s) . 0 means disabled

- VerificationType Verification Method

...

1. Standard Mode Default Value
2. Face/Fingerprint/Palm Print/Card + Password
3. Card+Face/Fingerprint/Palm Print/Password
4. Multiple People Attendance
5. Person and Identification Comparison
6. Card + Face/Fingerprint/Palm Print + Password
7. Card + Fingerprint/Palm Print + Face

8. Fingerprint/Palm Print + Face + Password
9. Fingerprint + Palm Print + Face
10. Palm Print + Face
11. Fingerprint + Face
12. Only use palm print
13. Only use fingerprint
14. Only use card
15. Only use password
16. Person and Identification Comparison Open + Registered Person
(Identification card + face + registered)

...

- **Weekly Time Period Format*

```
```json
{
 week1:"00:00-23:59", //Monday
 week2:"00:00-23:59",
 week3:"00:00-23:59",
 week4:"00:00-23:59",
 week5:"00:00-23:59",
 week6:"00:00-23:59",
 week7:"00:00-23:59" //Sunday
}
```
```

- Week1 means Monday
- Week2 means Tuesday
- Week3 means Wednesday
- Week4 means Thursday
- Week5 means Friday
- Week6 means Saturday
- Week7 means Sunday
- Each week field represents the time period of a day.
- Can set 8 sub-periods in one day, and the format is start time-end time/start time-end time/.....

...

"01:00-01:59/02:00-02:59/03:00-03:59/04:00-04:59/05:00-05:59/06:00-06:59/07:00-07:59/08:00-08:59"

The string above defines 8 sub-periods in a day. A maximum of 8 sub-periods can be defined in a day.

"01:00-01:59/02:00--02:59"

The above string defines two sub-periods of the day, and the other six periods are empty and invalid.

...

- If the week of the day is empty and the time period is not set, this field can be omitted

```

    json
    {
        //This period is defined only week1 and week7  Other time periods are
empty

        week1:"01:00-02:00",
        week7:"03:00-04:00"
    }
}

```

Elevator Function Parameters

| Field | Type | Required | Description |
|---------------|---------------------|----------|--|
| UseElevator | int | Yes | Elevator function switch, 1: on, 0: off |
| ElevatorPorts | [] Array of Objects | Yes | ElevatorPort Array of Objects
Define a list of elevator ports |

Elevator Port Object

| Field | Type | Required | Description |
|-------------|------|----------|---|
| Num | int | Yes | Elevator port number 1-64 |
| RelayType | int | Yes | Elevator relay (COM&NO normally closed, COM&NC normally closed) `<br>`Value range: 1. COM&NC normally closed (default value); 2. COM&NO normally closed |
| ReleaseTime | int | Yes | Output duration when unlocking. Maximum 65535 seconds. 0 means 0.5 seconds |
| TimingOpen | obj | Yes | Timed normally open function structure |

- timingOpen Timed normally open function

| Field | Type | Required | Description |
|-------|------|----------|--|
| Use | int | Yes | Function switch, 0-disable; 1-enable |
| Open | int | Yes | Automatic opening mode: `<br>`1. After passing the legal authentication, the door can be opened normally within the specified time period. `<br>`2. If the authorization is marked as the normally open privilege, the door can be opened normally within the specified time period after passing the authentication. `<br>`3. Automatic opening and closing, automatically opening and closing the door at the specified time |

| Timegroup | obj | Yes | Normally opening period, using weekly period structure

Alarm Parameter Alarm

| Field | Type | Required | Description |
|-------------------------------|--------|----------|---|
| : | : | : | : |
| FireAlarm | int | Yes | Fire alarm, 0, Disable; 1, Enable |
| DoorLongOpenAlarm | int | Yes | Door opening timeout alarm switch, 1: Enable, 0: Disable |
| DoorLongOpenTime | int | Yes | Door opening timeout, alarm will be triggered if the door is opened for more than this time 1-65535 (s) |
| DoorSensorAlarm | int | Yes | Door sensor alarm, 0, Disable; 1, Enable |
| DoorSensorAlarmTimegroup | class | Yes | Door sensor alarm or non-alarm period, weekly period format |
| BlacklistAlarm | int | Yes | Blacklist alarm, 0, Disable; 1, Enable |
| AntiDisassemblyAlarm | int | Yes | Tamper Alarm Function Switch, 0, Disable; 1, Enable |
| IllegalVerificationAlarm | int | Yes | Illegal verification alarm function, 0, Disable; 1, Enable |
| IllegalVerificationAlarmLimit | int | Yes | Illegal verification alarm function-number of illegal authentications, 1-255 |
| UseUserCloseAlarm | int | Yes | Allow the user to verify the alarm release switch, 0, Disable; 1, Enable |
| PasswordAlarm | int | Yes | Duress alarm password function, 0, Disable; 1, Enable |
| PasswordAlarm_Password | string | Yes | Duress alarm password. An alarm will occur when enter this password. The password only supports numbers and can contain 0. |
| PasswordAlarm_Mode | string | Yes | Working mode when duress alarm occurs ``1--Do not open the door, alarm output``2--Open the door, alarm output ``3--Lock the door, alarm, can only be unlocked by software`` |

Device Door Opening time period Timegroup

| Field | Type | Required | Description |
|------------|---------------------|----------|----------------------------|
| :----- | :----- | :-- | :----- |
| TimeGroups | [] Array of objects | Yes | Device Opening Time Period |

****Device opening door time period format description****

```json

//Parameter

```
TimeGroups:[
 {
 Num:1,
 week1:"00:00-23:59", //Monday

 week2:"00:00-23:59",
 week3:"00:00-23:59",
 week4:"00:00-23:59",
 week5:"00:00-23:59",
 week6:"00:00-23:59",
 week7:"00:00-23:59" //Sunday
 },
 {
 Num:2,
 week1:"01:00-01:59/02:00-02:59/01:00-01:59/02:00-02:59",
 ..
 week7:"00:00-00:00"
 },

 {
 Num:64,
 week1:"00:00-00:00",
 ..
 week7:"00:00-00:00"
 }
]
```

- The maximum num of device opening time periods is 64, which means the device has a maximum of 64 sets of opening time periods.

- week1 means Monday
- week1 means Monday
- week2 means Tuesday
- week3 means Wednesday
- week4 means Thursday
- week5 means Friday
- week6 means Saturday
- week7 mean Sunday
- Each week field indicates the time period of a day
- Can set 8 sub-periods in a day. The format is start time-end time/start time-end time/.....

```

...
"01:00-01:59/02:00-02:59/03:00-03:59/04:00-04:59/05:00-05:59/06:00-
06:59/07:00-07:59/08:00-08:59"
The string above defines 8 sub-periods in a day. A maximum of 8 sub-periods
can be defined in a day.

"01:00-01:59/02:00-02:59"
The above string defines two sub-periods of the day, and the other six periods
are empty and invalid.
...
- If the week is empty and the time period is not set, this field can be omitted
...
{
 Num:64, //This time period is defined only week1 and week7 other time
periods are empty.
 week1:"01:00-02:00",
 week7:"03:00-04:00"
}
...

```

### ### Device Holidays Holiday

| Field    | Type | Required | Description |
|----------|------|----------|-------------|
| Holidays | []   | Yes      | Holidays    |

```

```json
[
  {"Num":1,"Date":"2020-10-01","Type":1,"Cycle":1},
  {"Num":2,"Date":"2020-10-02","Type":2,"Cycle":0},
  ...
]
```

```

Holidays use an object array format, and each object contains two fields, num and date.

The device currently supports 360 groups of holidays

On holidays, it is forbidden to open the door (can set access level for holidays)

### \*\*Holiday object field description\*\*

| Field | Type   | Required | Description                                                            |
|-------|--------|----------|------------------------------------------------------------------------|
| Num   | int    | Yes      | The serial number of holidays, use when binding personnel access level |
| Date  | string | Yes      | Holiday Date Year-Month-Day Example:2020-10-01                         |



|       |     |    |                                                                                                                       |
|-------|-----|----|-----------------------------------------------------------------------------------------------------------------------|
| Type  | int | No | Holiday control range, 1--all day; 2--00:00-12:00 in the morning; 3--afternoon (12:00-23:59), the default value is 1; |
| Cycle | int | No | whether to cycle every year, 1--cycle every year; 0--no cycle; the default value is 0;                                |

### \*\*Alarm Clock\*\* AlarmClock

#### \*\*Up to 24 groups of alarm clock\*\*

| Field       | Type | Required | Description |
|-------------|------|----------|-------------|
| AlarmClocks | []   | Yes      | Alarm Clock |

```
```json
[
  {"Num":1,"Clock":"12:00","Times":10},
  {"Num":2,"Clock":"13:00","Times":10},
  {"Num":3,"Clock":"14:00","Times":10},
  ...
]
```
```

The alarm clock uses an object array format, and each object contains three fields: Num, Clock, and Times.

The device currently supports 24 sets of alarm clock

**\*\*Alarm clock object field description\*\***

| Field | Type | Required | Description                                                  |
|-------|------|----------|--------------------------------------------------------------|
| Num   | int  | Yes      | Alarm clock serial number, value range1-24                   |
| Clock | int  | Yes      | Alarm clock time point HHmm format example: 1230 means 12:30 |
| Times | int  | No       | Alarm ringing duration, unit: seconds, value range: 0-255    |

## \*\*API-Obtain the device function list\*\*

**\*\*Brief description:\*\***

- After logging in, call this interface to obtain the function list of the device and adjust the interface accordingly.

**\*\*Request URL:\*\***

- `http://Server IP:port/api/Device/FunctionList`

**\*\*Request Mode:\*\***

- POST

**\*\*verify Toekn\*\***

- verify

**\*\*Content-Type\*\***

- `application/json`

**\*\*Authorization\*\***

- Bearer token character string

### **\*\*Return example\*\***

````json`

`//Return function list supported by the device`

```
{
  "result": true ,
  "content": {
    //Body Temperature Detection

    "BodyTemperature": true,
    //Fingerprint

    "Fingerprint": true,
    //Palm Print

    "Palmvein": true,
    //Face Recognition

    "Face": true,
    //QR Code

    "QRCode": true,
    //Mask Detection

    "FaceMask": true,
    //Helmet Detection

    "SafetyHelmet": true,
    //Elevator

    "Lift": true,
    //Alarm Clock

    "AlarmClock": true,
    //excel export and import
  }
}
```

```

    "ExcelFile": true,
    //zip import

    "ZipFile": true,
    //Number of time period

    "TimeGreoup": true,
    //Wireless Network

    "WIFI": true,
    //HTTPClient v1
    "HTTPClient_v1": true,
    //HTTPClient v2
    "HTTPClient_v2": true,
    //MQTT
    "MQTT": true,
    //Yunzhu Platform

    "YZW": true,
    //Websocket v1
    "Websocket_v1": true,
    //Websocket v2
    "Websocket_v2": true,
    //Websocket Server
    "WebsocketServer": true,
    //Access control function--timed locking
    "TimingLocked": true,
    "SSH": true, //SSH Debugging Connection
    "Telnet": true, //Telnet Debugging Connection
    "FastRecognition": true, //Quick recognition, without displaying
recognized names and photos
    "RequestAuthorization": true, //Online authentication
    "RTSP" : false, //Does it have RTSP
    "TCPServer" : false, //Does it have TCPServer
    "RecognitionButon" : false, //Click to identify
    "DailyLimit" : false, //Daily recognition limit
    "OverdueRemind" : false, //Permission expiration prompt
    "DelayOpenDoorTime" : false, //Delayed unlocking
}
}

//Log Out

{
    "result": false,
    "errCode": 401,
    "error": "token verification failed"
}

...

```

API-Obtain device operating parameters

****Brief Description:****

- After logging in, call **this** interface to obtain the working status of the device

****Request URL: ****

- **http://server IP:port/api/Device/GetDetail**

****Request Mode:****

- POST

****Verify Toekn****

- verify

****Content-Type****

- application/json

****Authorization****

- Bearer token character string

**Request parameter**

| Field Name | Type | Required | Description |
|-----------------|------|----------|-----------------------------|
| SystemInfo | bool | No | Device Basic Info |
| Status | bool | No | Device Working Status |
| Language | bool | No | Region and language |
| UI | bool | No | Man-machine interaction |
| Storage | bool | No | Data storage |
| Face | bool | No | Face recognition |
| BodyTemperature | bool | No | Body temperature detection |
| NetworkServer | bool | No | Server Parameter |
| Network | bool | No | Device Network Parameter |
| Door | bool | No | Access Control Parameter |
| Elevator | bool | No | Elevator Function Parameter |
| Alarm | bool | No | Alarm Parameter |
| TimeGroups | bool | No | Device Opening Time Period |
| Holidays | bool | No | Device Holidays |
| AlarmClocks | bool | No | Alarm Clock |

~~~json

```
{
  "SystemInfo": true,
  "Status": true,
  "Language": true,
  "UI": true,
  "Storage": true,
  "Face": true,
```

```

    "BodyTemperature": true,
    "NetworkServer": true,
    "Network": true,
    "Door": true,
    "Elevator": true,
    "Alarm": true,
    "TimeGroups": true,
    "Holidays": true,
    "AlarmClocks": true
}

```

Return Parameter Description

| Field | Type | Required | Description |
|---|--------|----------|-------------|
| deviceId | string | Yes | Device ID |
| Other function parameters of the device | | | |

```

//Return on successful verification

{
  "result": true,
  "content": {

    //Alarm Parameters Alarm

    "FireAlarm": 1,
    "DoorLongOpenAlarm": 1,
    "DoorLongOpenTime": 60,
    "DoorSensorAlarm": 1,
    "DoorSensorAlarmTimegroup": {
      "week1": "00:00-00:50/01:00-01:50/02:00-02:50/03:00-03:50/04:00-04:50/05:00-05:50/06:00-06:50/07:00-07:50",
      "week2": "01:01-01:51/02:01-02:51/03:01-03:51/04:01-04:51/05:01-05:51/06:01-06:51/07:01-07:51/08:01-08:51",
      "week3": "02:02-02:52/03:02-03:52/04:02-04:52/05:02-05:52/06:02-06:52/07:02-07:52/08:02-08:52/09:02-09:52",
      "week4": "03:03-03:53/04:03-04:53/05:03-05:53/06:03-06:53/07:03-07:53/08:03-08:53/09:03-09:53/10:03-10:53",
      "week5": "04:04-04:54/05:04-05:54/06:04-06:54/07:04-07:54/08:04-08:54/09:04-09:54/10:04-10:54/11:04-11:54",
      "week6": "05:05-05:55/06:05-06:55/07:05-07:55/08:05-08:55/09:05-09:55/10:05-10:55/11:05-11:55/12:05-12:55",
      "week7": "06:06-06:56/07:06-07:56/08:06-08:56/09:06-09:56/10:06-10:56/11:06-11:56/12:06-12:56/13:06-13:56"
    },
    "BlacklistAlarm": 1,
    "AntiDisassemblyAlarm": 1,
    "IllegalVerificationAlarm": 1,
    "IllegalVerificationAlarmLimit": 5,
  }
}

```

```
"UseUserCloseAlarm": 1,
"PasswordAlarm": 1,
"PasswordAlarm_Password": "110110",
"PasswordAlarm_Mode": 2,
```

//Alarm Clock AlarmClock

```
"AlarmClocks": [{
    "Num": 1,
    "Clock": "00:01",
    "Times": 1
}, {
    "Num": 24,
    "Clock": "23:24",
    "Times": 24
}],
```

//BodyTemperature Detection BodyTemperature

```
"UseBodyTemperature": 1,
"UseFahrenheitDisplay": 1,
"TemperatureCompensate": -1.5,
"TemperatureAlarmThreshold": 37.5,
"TemperatureDisplay": 1,
```

//Access Control Parameters Door

```
"CardBytes": 4,
"AccessType": 0,
"WgFormat": 34,
"WGContent": 1,
"ReleaseTime": 3,
"FreeOpen": 1,
"OpenInterval": 30,
"OpenInterval_SaveRecord": 1,
"Relay": 1,
"ShortMessage": "Good morning",

"VerificationType": 1,
"OverdueRemind": 1,
"OverdueRemind_Day": 3,
"TimingOpen": 1,
"TimingOpen_mode": 3,
"TimingOpen_timegroup": {
    "Week1": "00:00-00:50/01:00-01:50/02:00-02:50/03:00-03:50/04:00-04:50/05:00-05:50/06:00-06:50/07:00-07:50",
    "Week7": "06:06-06:56/07:06-07:56/08:06-08:56/09:06-09:56/10:06-10:56/11:06-11:56/12:06-12:56/13:06-13:56"
```

```

},
"TimingLocked": 1,
"TimingLocked_timegroup": {
    "Week1": "00:00-00:50/01:00-01:50/02:00-02:50/03:00-03:50/04:00-04:50/05:00-05:50/06:00-06:50/07:00-07:50",
    "Week7": "06:06-06:56/07:06-07:56/08:06-08:56/09:06-09:56/10:06-10:56/11:06-11:56/12:06-12:56/13:06-13:56"
},
"VisitorRootPassword": "123456",
"MultiPerson": 1,

```

//Elevator Function Parameter Elevator

```

"UseElevator": 1,
"ElevatorPorts": [{
    "Num": 1,
    "RelayType": 0,
    "ReleaseTime": 3,
    "TimingOpen": {
        "Use": 1,
        "Open": 3,
        "Timegroup": {
            "Week1": "00:00-00:50/01:00-01:50/02:00-02:50/03:00-03:50/04:00-04:50/05:00-05:50/06:00-06:50/07:00-07:50",
            "Week7": "06:06-06:56/07:06-07:56/08:06-08:56/09:06-09:56/10:06-10:56/11:06-11:56/12:06-12:56/13:06-13:56"
        }
    }
}, {
    "Num": 64,
    "RelayType": 0,
    "ReleaseTime": 3,
    "TimingOpen": {
        "Use": 1,
        "Open": 3,
        "Timegroup": {
            "Week1": "00:00-00:50/01:00-01:50/02:00-02:50/03:00-03:50/04:00-04:50/05:00-05:50/06:00-06:50/07:00-07:50",
            "Week7": "06:06-06:56/07:06-07:56/08:06-08:56/09:06-09:56/10:06-10:56/11:06-11:56/12:06-12:56/13:06-13:56"
        }
    }
}],

```

//Face Recognition Face

```

"FaceIR": 1,
"FaceIRThreshold": 80,
"FaceDistance": 1,
"FaceThreshold": 65,
"FPComparison": 80,

```

```
"FaceMask": 0,  
"FaceMaskThreshold": 65,
```

//Device Holidays Holiday

```
"Holidays": [{  
    "Num": 1,  
    "Date": "2024-01-01",  
    "Type": 1,  
    "Cycle": 1  
}, {  
    "Num": 30,  
    "Date": "2024-01-30",  
    "Type": 1,  
    "Cycle": 1  
}],
```

//Region & Language Language

```
"Language": 1,  
"SystemTime": 1710489007,  
"TimeZone": 8,  
"Volume": 10,  
"Voice": 1,
```

//Device Networking Parameters Network

```
"ConnectPassword": "12345678",  
"UseWired": 1,  
"WiredDHCP": 1,  
"WiredIP": "192.168.1.150",  
"WiredIPMask": "255.255.255.0",  
"WiredGteway": "192.168.1.1",  
"WiredDNS": "8.8.8.8",  
"WiredMAC": "01-02-03-04-05-06",  
"UseWiFi": 1,  
"WIFIDHCP": 1,  
"WIFIIP": "192.168.1.150",  
"WIFIIPMask": "255.255.255.0",  
"WIFIGteway": "192.168.1.1",  
"WIFIDNS": "8.8.8.8",  
"WIFIMAC": "01-02-03-04-05-06",  
"WIFIAPName": "abcd",  
"WIFIAPPassword": "12345678",  
"UseWebPage": 1,  
"WebPagePort": 80,  
"WebPageUseSSL": 1,  
"UseUDP": 1,  
"UDPPort": 8101,  
"UseTelnet": 1,  
"TelnetPort": 23,  
"UseRTSP": 1,
```



```
"RTSPPort": 554,  
"RTSPUser": "admin",  
"RTSPPassword": "12345678",
```

//Server Parameters NetworkServer

```
"ServerAddress": "www.pc15.net",  
"ServerPort": 9003,  
"keepaliveTime": 30,  
"ServerProtocol": 1,  
"HttpRequestUseGZIP": 1,
```

//Device Working Status Status

```
"RunDays": 10,  
"FormatCount": 66,  
"WatchDogCount": 13,  
"BootTime": 1710489007,  
"RelayStatus": 1,  
"KeepOpenStatus": 1,  
"DoorSensorStatus": 1,  
"LockDoorStatus": 1,  
"AlarmStatus": "fire,blacklist",
```

//Data Storage Storage

```
"RecordAutoCycle": 1,  
"SaveUnregistered": 1,  
"SaveRecordPicture": 1,  
"PeopleStorageInfo": {  
  "Person": {  
    "Max": 10000,  
    "Current": 100  
  },  
  "Face": {  
    "Max": 20000,  
    "Current": 200  
  },  
  "Card": {  
    "Max": 30000,  
    "Current": 300  
  },  
  "Fingerprint": {  
    "Max": 40000,  
    "Current": 400  
  },  
  "PalmVein": {  
    "Max": 50000,  
    "Current": 500  
  },  
}
```

```

    "Pasword": {
        "Max": 60000,
        "Current": 600
    },
    "Admin": {
        "Max": 70000,
        "Current": 700
    }
},
"RecordStorageInfo": {
    "VerifyRecord": {
        "Max": 80000,
        "Current": 800
    },
    "DoorRecord": {
        "Max": 90000,
        "Current": 900
    },
    "SystemRecord": {
        "Max": 11000,
        "Current": 110
    },
    "RecordPhoto": {
        "Max": 12000,
        "Current": 120
    }
}
},

```

//Device Basic Info SystemInfo

```

"DeviceSN": "FC-8190H12345678",
"DeviceName": "FC-8190H",
"FirmwareVersion": "8.23",
"FingerprintVersion": "8.23",
"FaceVersion": "6.01",
"Manufacturer": "Guangzhou xxxxxxxxxxx Technology Co., Ltd.",
"ManufacturerPhone": "020-12345678",
"Website": "www.abc123.com",
"ProductionDate": "2024-01-01",
"OEMText": "abcdefg12345 Hi",
"AutoRestart": 0,
"AutoRestartTime": 0,

```

//Device Opening Time Zone Timegroup

```

"TimeGroups": [{
    "Num": 1,
    "week1": "00:00-00:50/01:00-01:50/02:00-02:50/03:00-03:50/04:00-04:50/05:00-05:50/06:00-06:50/07:00-07:50",
    "week7": "06:06-06:56/07:06-07:56/08:06-08:56/09:06-09:56/10:06-10:56/11:06-11:56/12:06-12:56/13:06-13:56"
}, {

```

```

        "Num": 64,
        "week1": "00:00-00:50/01:00-01:50/02:00-02:50/03:00-03:50/04:00-04:50/05:00-05:50/06:00-06:50/07:00-07:50",
        "week7": "06:06-06:56/07:06-07:56/08:06-08:56/09:06-09:56/10:06-10:56/11:06-11:56/12:06-12:56/13:06-13:56"
    }],

    //Man-machine interaction UI
    "DisplayBrightness": 10,
    "MenuPassword": "123456",
    "ShowIR": 1,
    "ShowPersonPhoto": 1,
    "PlayPersonName": 1,
    "RecognitionButon": 1,
    "UnregisteredWarn": 1,
    "ShowPersonName": 1,
    "FillLight": 2

}

}

```

```

//Log out
{
    "result": false,
    "errCode": 401,
    "error": "token verification failed"
}

```

API-update device working parameters

Brief Description::

- When the user modifies the device working parameters, use this interface to upload the modified working parameters

Request URL:

- <http://Server IP:port/api/Device/UpdateParameter>

Request Mode

- POST

Verify Toekn

- Verify

Content-Type

- application/json

Authorization

- Bearer token character string

Request Parameters

- Device working parameter fields that need to be updated

Request example

```
{
  "DeviceSN": "FC-8190H12345678",
  "DeviceName": "FC-8190H",
  "Manufacturer": "Manufacturer Manufacturer Manufacturer",
  "ManufacturerPhone": "020-12345678",
  "Website": "www.abc123.com",
  "ProductionDate": "2024-01-01",
  "OEMText": "OEMText OEMText OEMText",
  "AutoRestartTime": "23:59",
  "RecordAutoCycle": 1,
  "SaveUnregistered": 1,
  "SaveRecordPicture": 1,
  "AutoRestart": 0
}
```

Return example

```
//Update success return

{ "result": true , "content": "OK" }

//Log out

{
  "result": false,
  "errCode": 401,
  "error": "token verification failed"
}

//Parameter error return

{
  "result": false,
  "errCode": 1,
  "error": "parameter error: ..... "
}
```

API-send device control command

Brief Description :

- Send when the user clicks on a remote control command on the device page

Request URL:

- <http://Server> IP:port/api/Device/Remote

Request Mode

- POST

Verify Token

- Verify

Content-Type

- application/json

Authorization

- Bearer token character string

Request Parameters

| Command | Type | Description | Required |
|------------|------|-------------------------------------|----------|
| OpenDoor | bool | Remote door opening | No |
| CloseDoor | bool | Remote door closing | No |
| KeepOpen | bool | Remote normally open | No |
| LockDoor | bool | Remote lock | No |
| UnlockDoor | bool | Remote unlock | No |
| FireAlarm | bool | Let the device have a fire alarm | No |
| CloseAlarm | bool | Disable alarm | No |
| Restart | bool | Restart the device | No |
| Recover | bool | Reset the device to factory default | No |

Request example

```
//Remote door opening
{
  "OpenDoor": true
}
```

Return example

```
//Return when verification succeeds
{
  "result": true
}

//Log out
{
```

```

"result": false,
"errCode": 401,
"error": "token verification failed"

}

//Parameter error return

{
  "result": false,
  "errCode": 1,
  "error": "no action to be performed"
}

```

API-update device firmware

Brief Description:

- When the user is on the device page, called when uploading new firmware

Request URL:

- <http://Server IP:port/api/Device/UploadSoftware>

Request Mode

- POST

Verify Toekn

- Verify

Content-Type

- multipart/form-data

Authorization

- Bearer token character string

Request Parameters

Request Parameters

| Field | Type | Length | Required | Description |
|--------------|--------------------------|--------|----------|---|
| SoftwareMD5 | string | 30 | Yes | MD5 value of the firmware |
| SoftwareFile | application/octet-stream | 30 | Yes | Firmware File
This field must contain
Content-Type:
application/octet-stream |

.Request example

Request example

```
POST /note/insertNoteFace HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-----
-506873351428002157394455
Content-Length: 17842

-----506873351428002157394455
Content-Disposition: form-data; name="softwareMD5"

abcdefghijklmnopqrstuvwxyz
-----506873351428002157394455
Content-Disposition: form-data; name="softwareFile"; filename="softwareFile.pkg"
Content-Type: application/octet-stream

*****jpeg file binary content*****

-----506873351428002157394455--
```

Return example

```
//Return when verification succeeds

{
  "result": true
}
```

Personnel Management

API-obtain a list of users

Brief Description:

- When querying users on the UI, call this interface

Request URL:

- <http://Server IP:port/api/People/Search>

Request mode:

- POST

Authorization

- Bearer token character string

Request required parameters

| Field | Type | Length | Required | Description |
|-----------|--------|--------|----------|-------------------------------------|
| PageIndex | uint32 | | Yes | Current page index, starting from 1 |

| Field | Type | Length | Required | Description |
|----------|--------|--------|----------|-------------------------------------|
| PageSize | uint32 | | Yes | Number of people per page requested |

Request optional parameters

- Optional parameters for filtering queries

| Field | Type | Length | Required | Description |
|---------------|--------|--------|--|-------------|
| UserID | string | No | User ID (characters < 32 bits) | |
| Name | string | No | User Name (characters < 32 bits) | |
| Job | string | No | Position (characters < 32 bits) | |
| Department | string | No | Department (characters < 32 bits) | |
| AccessType | int | No | Role 0--Staff;1--Admin;2--Blacklist | |
| Timegroup | int | No | Opening time group 1-64 | |
| Photo | int | No | Are there any photos? 1--Yes;0--No | |
| CardNum | string | No | IC card number pure number
maximum support uint64 | |
| IdentityCard | string | No | Identity Card Number (Can be empty) | |
| Fingerprint | int | No | Are there any fingerprints? 1--Yes;0--No | |
| Palmprint | int | No | Are there any palm prints? 1--Yes;0--No | |
| OrderByColumn | string | No | The column to sort by, the default is UserID
The optional contains UserID,Name,Department, Job, CardNum | |
| OrderByType | string | No | ASC or DESC Sort ordering, ascending or descending order
Default is ASC | |

Request example

```
{
  "PageIndex":1,
  "PageSize":20,
  "UserName":"wang" //This field indicates that you are searching for people
  whose usernames contain "wang"
}
```

Return example

| |
|--|
| |
|--|


```

{
  "result": true,
  "errCode": 0,
  "msg": "",
  "content": {
    "TotalCount": 2000, //The total number of users queried

    "PageIndex": 1, //Current page index

    "PageSize": 20, //Number of users currently on the page

    "DataList": [ //Returned User List

      {
        "UserID": "1",
        "Name": "Test Name1",
        "Job": "",
        "Department": "Sales Department",
        "Password": "",
        "CardNum": "0",
        "AccessType": 0,
        "ExpirationDate": 4102415940,
        "OpenTimes": 65535,
        "KeepOpen": 0,
        "Timegroup": 1,
        "FaceFeature": 0,
        "photo": "",
        "Fingerprint": 0,
        "Palmprint": 0
      },
      {
        "UserID": "10",
        "Name": "11",
        "Job": "None",
        "Department": "Default Department",

        "Password": "",
        "CardNum": "0",
        "AccessType": 0,
        "ExpirationDate": 1909065540,
        "OpenTimes": 65535,
        "KeepOpen": 0,
        "Timegroup": 1,
        "FaceFeature": 1,
        "photo": "/data/attend_data/photo/frame_1711509862529292.jpg",
        "Fingerprint": 0,
        "Palmprint": 0
      }
    ]
  }
}

```

API-obtain personnel details

Brief Description::

- Call this interface when modifying a user on the UI

Request URL:

- <http://Server IP:port/api/People/GetDetail>

Request Mode:

- Post

Authorization

- Bearer token character string

Parameter:

| Field | Type | Length | Required | Description |
|-------------|--------|--------|----------|--|
| UserID | string | 32 | Yes | The User ID to be obtained |
| PhotoBase64 | int | 1 | No | Request to return the base64 encoding of the photo |

Request example

```
{
  "UserID": "3007465",
}
```

Return example

```
{
  "result": true,
  "content": {
    "UserID": "3",
    "Name": "888888",
    "Job": "Developing",
    "Department": "Sales Department",
    "IdentityCard": "",
    "Attachment": "",
    "PhotoMD5": "613D870CA99EDF074BEE4387BAB09070",
    "PhotoLen": 55020,
    "Password": "2222",
    "CardNum": "6666",
    "AccessType": 0,
    "ExpirationDate": 0,
    "OpenTimes": 65535,
    "KeepOpen": 1,
    "Timegroup": 6,
    "Holidays": "1,3,9,10,11,17,21,25,27,30",
    "Elevators": "1,2,3,4,5",
    "FaceFeature": "AxwtAQEbRcY3HDIu9A/9AP/46Q7dT ....",
    "Fingerprints": [],
  }
}
```

```
    "Palmveins": [],  
    "Photo": "/9//4AAQSkZJRgABAQAAQABAA//4g/YS....."  
  }  
}
```

API-Obtain new user ID

Brief Description:

- Call this interface to get the available person ID before adding people on the UI

Request URL:

- <http://Server> IP:port/api/People/GetNewID

Request Mode:

- Post

Authorization

- Bearer token character string

Return example

```
{  
  "result": true,  
  "content": {  
    "NewUserID": 3  
  }  
}
```

##

API-Add new user

Brief Description:

- This interface is called when adding a new person on the UI. If a person with the same number already exists, it will be overwritten.

Request URL:

- <http://Server> IP:port/api/People/New

Request Mode:

- POST

Content-Type

- multipart/form-data

Authorization

- Bearer token character string

Parameter:

| Field | Type | Length | Required | Description |
|-------|------|--------|----------|-------------|
|-------|------|--------|----------|-------------|

| Field | Type | Length | Required | Description |
|------------|--------|--------|----------|---|
| Peoplejson | string | | Yes | Personnel details, json string |
| Photo | file | 500KB | No | This parameter is required only when the person has a photo |

Peoplejson Personnel Data Format

| Parameter | Type | Required? | Description |
|----------------|--------|-----------|---|
| UserID | string | Yes | User ID (number, maximum value 4294967295, type UINT32) |
| Name | string | No | User Name(Character < 64 bits) |
| Job | string | No | Position(character < 64 bits) |
| Department | string | No | Department(character < 64 bits)) |
| IdentityCard | string | No | Identity Card Number (Can be empty) (characters < 64) |
| Attachment | string | No | Other users info (characters < 200) |
| Photo | string | | User photo, can be empty
Support base64 encoding of photo files or file upload |
| PhotoMD5 | string | No | MD5 hexadecimal string format of the photo (character = 32 bits) |
| PhotoLen | int | No | The maximum supported photo length is 400KB |
| Password | string | No | Password, pure numbers, length: (0 / 4-8) |
| CardNum | string | No | Card number (number, maximum value 18446744073709551615, type UINT62) |
| QRCode | string | No | User QR code info (characters <128 bits) |
| AccessType | int | No | Role 0--Staff; 1--Admin; 2--Blacklist |
| ExpirationDate | uint32 | No | Access Level Expiration Date
unix Timestamp Second 0 means no time limit |
| OpenTimes | int | No | Opening Times 0-65535;
65535--no limit, 0--no entry |
| KeepOpen | int | No | Is the normally open card? 1--Yes; 0--No |
| Timegroup | int | No | Door opening time group 1-64; 0--restricted |

| Parameter | Type | Required? | Description |
|----------------|--------|-----------|---|
| Holidays | string | No | Holidays Limit
Comma Separated:1,2,3,4,5 |
| Fingerprints | [] | No | Fingerprint object |
| Palmveins | [] | No | Palmprint object |
| Elevators | string | No | Elevator Port Access Level Group
Comma Separated:1,2,3,4,5 |
| FaceFeature | string | No | Face feature code base64 encoded face
feature code |
| FaceFeatureMD5 | string | No | MD5 value of the face feature code HEX
character string format |

Characters refer to bytes. A Chinese character occupies 3-4 bytes (utf8 encoding), and an English character occupies 1 byte.

Holidays Holidays

- Empty string or no such field means no holiday restriction
- Number the specific restricted holidays,Comma Separated:1,2,3,4,5
- ○ means all holidays are subject to restrictions

- **Fingerprint Fingerprint object**

| Serial Number | Type | Required | Description |
|---------------|--------|----------|---|
| Num | int | Yes | Fingerprint index number |
| Data | string | Yes | Base64 encoded value of the fingerprint feature
code |
| MD5 | string | No | MD5 value of the feature code |

```
[ //Structure Example
  {
    Num: 1,
    Data: "asdfasdfasfdasdfafsd",
  },
  {
    Num: 2,
    Data: "asdfasdfasfdasdfafsd",
  }
]
```

- **Palmvein Palm vein object**

| Serial Number | Type | Required | Description |
|---------------|--------|----------|--|
| Num | int | Yes | Palm vein index number |
| Data | string | Yes | base64 encoded value of palmprint feature code |
| MD5 | string | No | MD5 value of the feature code |

```
[ //Structure Example
  {
    Num: 1,
    Data: "asdfasdfasfdasdfafsd",
  },
  {
    Num: 2,
    Data: "asdfasdfasfdasdfafsd",
  }
]
```

- **Elevator Elevator Permission**

```
//Indicates that this person has elevator access permission to floors 1-5
[
  1,2,3,4,5
]
//Indicates that this person does not have elevator access level

[

]
//Indicates that this person only has access to one elevator on the 10th floor

[
  10
]
```

Request example

```
POST /api/People/New HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-----
-506873351428002157394455
Content-Length: 17842

-----506873351428002157394455
Content-Disposition: form-data; name="PeopleJson"
```

```

{
  "UserID": 3007465,
  "UserName": "User 3007465",
  "Job": "Staff",
  "Password": "123456",
  "PhotoMD5": "rgssPASg5/rG4HYlUwb6CA==",
  "AccessType": 0,
  "Timegroup": 2
}
-----506873351428002157394455
Content-Disposition: form-data; name="Photo"; filename="Postman_file.jpg"
Content-Type: image/jpeg

****jpeg file binary content****

-----506873351428002157394455--

```

Return parameter

| Field | Type | Description |
|------------------|--------|--|
| result | bool | Operation result, true means success and false means failure |
| errCode | int | Error code, including common code
2--System busy, please try again later
3--Request parameter error
4--Request payload is too large and exceeds the limit
10--JSON parsing failed
11--Personnel parameter verification failed
12--Photo md5 verification failed
13--Photo size verification failed
20--Face feature code duplicated
21--Fingerprint feature code duplicated
22--Palm vein feature code duplicated
23--Card number duplicated
24--Personnel storage is full
25--Error in querying data
26--Error in saving personnel photos
27--Face feature code format error
28--Fingerprint feature code format error
29--Palm vein feature code format error
30--Photo cannot be recognized
31--Error in saving fingerprint feature code
32--Error in saving palm vein feature code |
| RepetitionUserID | string | Duplicate user ID
Can be empty, only returned when the face or card is repeated
Indicates who the card or face is repeated with |

Return example

```
//Added successfully
//Add

{
  "result": true
}
```

API-Delete user

Brief Description:

- Call this interface when modifying a user on the UI

Request URL:

- <http://Server IP:port/api/People/Delete>

Request Mode:

- POST

Content-Type

- application/json

Authorization

- Bearer token character string

Parameter:

| Field | Type | Length | Required | Description |
|-----------|------|--------|----------|--|
| UserIDs | [] | | No | User ID list of the user to be deleted |
| DeleteALL | int | | No | Clear all users, 1- means clear all users; 0- means only delete the specified user |

Request example

```
//Delete the specified user
{
  "UserIDs": [
    1,
    2,
    3,
    4
  ]
}
```

```
//Delete all users

{
  "DeleteALL": 1
}
```

Return example


```
{  
  "result": true  
}
```

API-Register personnel credentials on the device

Brief description:

- Call this interface when modifying personnel on the UI

Request URL:

- <http://Server> IP:port/api/**People/EnrollUserMediaData**

Request method:

- POST

Content-Type

- application/json

Authorization

- Bearer token character string

Parameter:

| Field | Type | Length | Required | Description |
|-------------|--------|--------|----------|--|
| EnrollType | string | | Yes | Registration Type |
| UserID | int | | Yes | User ID of personnel requiring registration credentials |
| EnrollIndex | int | | Yes | Registration certificate number required
For fingerprints and palm veins
Fingerprint number value range 0-9
Palm vein value range 1-2 |

EnrollType Description

| Value | Description |
|-------|----------------------|
| Face | Register Face |
| Card | Register Card |
| PIN | Register Password |
| FP | Register Fingerprint |
| Palm | Register Palm Vein |

Request Example

```
//Register Face
{
  "EnrollType": "Face",
  "UserID": 123,
  "EnrollIndex": 1
}
```

```
//Delete all personnel
{
  "EnrollType": "FP",
  "UserID": 123,
  "EnrollIndex": 1
}
```

Return value parameter

| Field | Type | Length | Required | Description |
|------------|------|--------|----------|---|
| UserID | int | | Yes | Personnel user number for registration certificate |
| Result | int | | Yes | Registration certificate result
1, Successful registration;
2, User cancels operation;
3--Duplicate registration information |
| Repetition | int | | No | Duplicate user numbers when registering credentials |

Json

```
//Registration success returned
{
  "result": true,
  "content": {
    "UserID": 123,
    "Result": 1
  }
}
```

Department Management

API-Obtain department list

Brief Description:

- Call this interface when querying departments on the UI

Request URL:

- <http://Server IP:port/api/Department/Search>

Request Mode:

- POST

Authorization

- Bearer token character string

Request required parameters

| Field | Type | Length | Required | Description |
|------------------------------------|--------|--------|----------|--|
| PageIndex | uint32 | | Yes | Index of the current page, starting with 1 |
| PageSize | uint32 | | Yes | Number of departments per page requested |
| Request optional parameters | | | | |

- Optional parameters for filtering queries

| Field | Type | Length | Required | Description |
|--------|--------|--------|---------------------------------------|-------------|
| DeptID | int | No | Department ID 1-1024 | |
| Name | string | No | Department Name(Characters < 20 bits) | |

Request example

```
{
  "PageIndex":1,
  "PageSize":20,
  "Name":"Development Department"
}
```

Return example

```
{
  "result": true,
  "errCode":0,
  "msg": "",
  "content": {
    "TotalCount": 20, //Total number of departments queried
    "PageIndex":1, //Current page index
    "PageSize":20, //Current page department number
  }
}
```

```

    "DataList":[//List of returned departments
      {
        "DeptID": "1",
        "Name": "Department1",

      },
      {
        "UserID": "2",
        "Name": "Department2"

      }
    ]
  }
}

```

API-Obtain new department ID details

Brief Description:

- Call this interface to obtain the available department ID before adding a department on the UI.

请求URL:

Request URL:

- <http://Server IP:port/api/Department/GetNewID>

Request Mode:

- Post

Authorization

- Bearer token character string

Return example

```

{
  "result": true,
  "content": {
    "NewDeptID": 3
  }
}

```

**API-Add new/Modify department*

Brief Description:

- When calling this interface to add a new department on the UI, if the department with the same number already exists, it will be overwritten.

Request URL:

- <http://Server IP:port/api/Department/New>

Request Mode:

- POST

Content-Type

- multipart/form-data

Authorization

- Bearer token character string

Parameter:

Department Format

| Parameter Name | Type | Required | Description |
|----------------|--------|----------|----------------------------------|
| DeptID | int | Yes | Department ID |
| Name | string | No | User Name (characters < 32 bits) |

Request example

```
{
  "DeptID": 1,
  "Name": "Department1"
}
```

Return parameter

| Field | Type | Description |
|---------|------|--|
| result | bool | Operation result, true means success, false means failure |
| errCode | int | Error code, including common code
2--System busy, please try again later
3--Request parameter error
4--Department name duplicated |

Return example

```
//Added successfully

{
  "result": true
}
```

API-Delete department

Brief Description:

- Call this interface when deleting a department on the UI.

Request URL:

- <http://Server IP:port/api/Department/Delete>

Request Mode:

- POST

Content-Type

- application/json

Authorization

- Bearer token character string

Parameter:

| Field | Type | Length | Required | Description |
|-----------|------|--------|----------|--|
| DeptIDs | [] | | No | ID list of the department to be deleted |
| DeleteALL | int | | No | Clear all departments, 1- means clear all departments; 0- means only delete the specified department |

Request example

```
//Delete specified user
```

```
{
  "DeptIDs": [
    1,
    2,
    3,
    4
  ]
}
```

```
//Delete all users
```

```
{
  "DeleteALL": 1
}
```

Return example

```
{
  "result": true
}
```

Records Management

API-Query punch in records

Brief Description:

- Call this interface when querying records&events on the UI

Request URL:

- <http://Server IP:port/api/Record/Identify/Search>

Request Mode:

- POST

Authorization

- Bearer token character string

Request required parameters

| Field | Type | Length | Required | Description |
|-----------|--------|--------|----------|---|
| PageIndex | uint32 | | Yes | Current page index, starting from 1 |
| PageSize | uint32 | | Yes | Request the number of records per page 1-1000 |
| BeginDate | uint32 | | Yes | Start time Unix timestamp milliseconds |
| EndDate | uint32 | | Yes | End time Unix timestamp milliseconds |

Request optional parameter

- Optional parameters for filtering queries

| Field | Type | Required | Description |
|---------------|--------|----------|---|
| UserID | string | No | User ID (characters < 32 bits) |
| Name | string | No | User Name (characters < 32 bits) |
| Department | string | No | Department (characters < 32 bits) |
| Job | string | No | Position (characters < 32 bits) |
| CardNum | string | No | IC card number Pure numbers Maximum supported: uint64 |
| IdentityCard | string | No | Identity Card Number (Can be empty) |
| RecordTypes | string | No | List of record types Comma separated, example: 1,2,3 |
| PhotoBase64 | int | No | Whether the image uses Base64 encoding, 1 - use Base64, 0 - do not use; |
| RecordID | int | No | Query records that are greater than or equal to a certain record number. The operator is >= |
| OrderByColumn | string | No | Sorted columns, default is RecordDate
The alternatives are
RecordID,RecordDate,UserID,Name,Department, Job, CardNum |
| OrderByType | string | No | ASC or DESC Sort order, ascending or descending order
Default is ASC |

Request example

```
{
  "PageIndex":1,
  "PageSize":20,
  "UserName":"wang">//This field indicates that the record with the username
  containing wang is searched
}
```

Return example

```
{
  "result": true,
  "errCode":0,
  "msg": "",
  "content": {
    "TotalCount": 2000,//The total number of records queried
    "PageIndex":1,//Current page index
    "PageSize":20,//Number of current page records
    "DataList":[//List of returned records

      {
        "RecordID": 1000,//Record ID,each record should have an ID

        "UserID": "1",
        "Name": "Test Name 1",

        "IdentityCard": "",
        "Job": "",
        "Department": "Sales Department",

        "CardNum": "0",
        "RecordType": 1,
        "IsEntry": 0,//Is it an entry record?

        "RecordDate": 1710467919,
        "BodyTemp": 0,//Body Temperature

        "PhotoLen": 17842,
        "Photo": "/RecordImage/1000.jpg", //The address of the on-site
        photo
      }
    ]
  }
}
```

Record Field Description

| Field | Type | Length | Description |
|----------|------|--------|----------------------|
| RecordID | long | 10 | Record serial number |
| UserID | long | 20 | User number, user ID |

| Field | Type | Length | Description |
|--------------|--------|--------|--|
| Name | string | 30 | User Name |
| IdentityCard | string | 30 | Identity Card |
| Job | string | 30 | Position |
| Department | string | 30 | Department |
| CardNum | string | 10 | Card Number |
| QRCode | string | 128 | QR Code |
| RecordType | int | 5 | Event Type |
| IsEntry | int | 1 | Whether to enter, 1 means entering, 0 means leaving |
| RecordDate | string | 20 | Record time unix timestamp Second |
| BodyTemp | float | 5 | Body Temperature Measurement |
| PhotoLen | int | 10 | Image file length 0 indicates that there is no image |
| Photo | string | | Photo address or base64 |

RecordType Event Type

| Value | Explanation |
|-------|--|
| 1 | Card |
| 2 | Fingerprint |
| 3 | Face |
| 4 | Card + Fingerprint |
| 5 | Face + Fingerprint |
| 6 | Card + Face |
| 7 | Card + Password |
| 8 | Face + Password |
| 9 | Fingerprint + Password |
| 10 | Password verification User ID+Password |
| 11 | Card + Fingerprint + Password |
| 12 | Card + Face + Password |
| 13 | Fingerprint + Face + Password |
| 14 | Card + Fingerprint + Face |
| 15 | Repeat Verification |

| Value | Explanation |
|-------|---|
| 16 | Expiration date has expired |
| 17 | Opening period expired |
| 18 | Not open on holidays |
| 19 | Unregistered user |
| 20 | Detection lock |
| 21 | The valid times have been used up |
| 22 | Verify when locked, prohibit opening the door |
| 23 | Report lost card |
| 24 | Blacklist card |
| 25 | Open door without verification -- When press the fingerprint, the user number is 0, and when swipe the card, the user number is the card number |
| 26 | Disable card swiping verification -- When card swiping is disabled in [Authorization Method] |
| 27 | Disable fingerprint verification -- When fingerprint is disabled in [Authorization Method] |
| 28 | Controller expired |
| 29 | Verification passed—validity period is about to expire |
| 30 | Abnormal body temperature, entry denied |
| 31 | Visitor password to open the door |
| 32 | Scan the dynamic QR code to open the door |
| 33 | Add a new user in the device menu |
| 34 | Modify user in the device menu |
| 35 | Delete user in the device menu |
| 36 | Palmprint |
| 37 | Card + Palmprint + Face |
| 38 | Palmprint + Password |
| 39 | Card + Palmprint |
| 40 | Face + Palmprint |
| 41 | Card + Palmprint + Password |
| 42 | Palmprint + Face + Password |

| Value | Explanation |
|-------|---|
| 43 | Fingerprint + Palmprint + Face |
| 44 | Combined verification--wait for the next person |
| 45 | Combination verification failed |
| 46 | Combination verification completed |
| 47 | Person and identity card comparison |

API-Query door sensor records

Brief Description:

- Call this interface when querying record events on the UI

Request URL:

- <http://Server IP:port/api/Record/DoorSensor/Search>

Request Mode:

- POST

Authorization

- Bearer token character string

Request required parameters

| Field | Type | Length | Required | Description |
|-----------|--------|--------|----------|--|
| PageIndex | uint32 | | Yes | Current page index, starting from 1 |
| PageSize | uint32 | | Yes | Request the number of records per page |
| BeginDate | uint32 | | Yes | Start time unix timestamp milliseconds |
| EndDate | uint32 | | Yes | End time unix timestamp milliseconds |

Request optional parameter

- Optional parameters for filtering queries

| Field | Type | Required | Description |
|---------------|--------|----------|---|
| RecordTypes | string | No | List of record types, comma-separated, example: 1,2,3 |
| RecordID | int | No | Query records larger than a certain record number |
| OrderByColumn | string | No | Sorted column,default is RecordDate
The alternatives are RecordID,RecordDate |
| OrderByType | string | No | ASC or DESC Sort order, ascending or descending order
The default is ASC |

Request example

```
{
  "PageIndex":1,
  "PageSize":20
}
```

return example

```
{
  "result": true,
  "errCode":0,
  "msg": "",
  "content": {
    "TotalCount": 2000, //The total number of records queried
    "PageIndex":1, //Current page index
    "PageSize":20, //Number of current page records
    "DataList": [ //List of returned records
      {
        "RecordID": 1000, //record ID, each record should have an ID
        "RecordType": 1,
        "RecordDate": 1710467919,
      }
    ]
  }
}
```

Record field description

| Field | Type | Length | Description |
|------------|--------|--------|---------------------------------------|
| RecordID | long | 10 | Record Number |
| RecordType | int | 5 | Events Type |
| RecordDate | string | 20 | Recording time unix timestamp seconds |

**RecordType Event Type

| Value | Explanation |
|-------|--|
| 1 | Door Sensor-open the door |
| 2 | Door Sensor-close the door |
| 3 | Enter the door status sensor alarm detection state |
| 4 | Exit the door status sensor alarm detection state |
| 5 | Door open |
| 6 | Open the door with a button |
| 7 | The door is locked when the button opens |

| Value | Explanation |
|-------|---|
| 8 | The controller has expired when the button opens the door |

API-Query system record

Brief Description:

-Call this interface when querying record events on the UI

Request UR:

- <http://Server IP:port/api/Record/System/Search>

Request Mode:

- POST

Authorization

- Bearer token character string

Request required parameter

| Field | Type | Length | Required | Description |
|-----------|--------|--------|----------|--|
| PageIndex | uint32 | | Yes | Index of the current page, starting with 1 |
| PageSize | uint32 | | Yes | Request the number of records per page |
| BeginDate | uint32 | | Yes | Start Time unix timestamp in milliseconds |
| EndDate | uint32 | | Yes | End Time unix timestamp in milliseconds |

Request optional parameter

- Optional parameters for filter queries

| Field | Type | Required | Description |
|---------------|--------|----------|---|
| RecordTypes | string | No | List of record types,comma separated, example: 1,2,3 |
| RecordID | int | No | Query records larger than a certain record number |
| OrderByColumn | string | No | Sorted column,default is RecordDate
The alternatives are RecordID,RecordDate |
| OrderByType | string | No | ASC or DESC Sort order, ascending or descending order
The default is ASC |

Request example

```
{
  "PageIndex":1,
  "PageSize":20
}
```

Return example

```
{
  "result": true,
  "errCode":0,
  "msg": "",
  "content": {
    "TotalCount": 2000, //The total number of records queried
    "PageIndex":1, //Current page index
    "PageSize":20, //Number of current page records

    "DataList":[ //List of returned records
      {
        "RecordID": 1000, //Record ID, Each record should have an ID

        "RecordType": 1,
        "RecordDate": 1710467919,
      }
    ]
  }
}
```

Record Field Description

| Field | Type | Length | Description |
|------------|--------|--------|------------------------------------|
| RecordID | long | 10 | Record serial number |
| RecordType | int | 5 | Event Type |
| RecordDate | string | 20 | Record time unix timestamp seconds |

RecordType Event Type

| Value | Explanation |
|-------|---|
| 1 | Software Opening |
| 2 | Software Closing |
| 3 | Software Normally Open |
| 4 | The controller automatically enters the normally open state |
| 5 | Controller automatically closes the door |
| 6 | Long press the exit button to open normally |

| Value | Explanation |
|-------|--|
| 7 | Long press the exit button to close normally |
| 8 | Software locked |
| 9 | Software unlocked |
| 10 | Controller timed lock - automatically lock when the time comes |
| 11 | Controller timed lock - automatically unlock when the time comes |
| 12 | Alarm--Locked |
| 13 | Alarm--Unlocked |
| 14 | Illegal Authentication Alarm |
| 15 | Door Sensor Alarm |
| 16 | Duress Alarm |
| 17 | Door Opening Timeout Alarm |
| 18 | Blacklist Alarm |
| 19 | Fire Alarm |
| 20 | Tamper Alarm |
| 21 | Remove illegal authentication alarm |
| 22 | Remove door sensor alarm |
| 23 | Remove duress alarm |
| 24 | Remove door opening timeout alarm |
| 25 | Remove blacklist alarm |
| 26 | Remove fire alarm |
| 27 | Remove tamper alarm |
| 28 | System Power On |
| 29 | System error reset (watchdog) |
| 30 | Device format record |
| 31 | Card reader connected reversely |
| 32 | The card reader line is not connected properly |
| 33 | Unrecognized card reader |
| 34 | Network cable disconnected |
| 35 | Network cable is inserted |

| Value | Explanation |
|-------|--|
| 36 | WIFI connected |
| 37 | WIFI disconnected |
| 38 | Bluetooth door opening |
| 39 | Roll Call Timeout |
| 40 | Clear all users in the device menu |
| 41 | Back up users to USB flash drive in the device menu |
| 42 | Import users from USB flash drive in the device menu |
| 43 | Indoor unit remote door opening |
| 44 | Delete all records |
| 45 | Delete all users |

API-Clear all records

Brief Description:

- Call this interface, when clear all records
- Request URL:**
- <http://Server IP:port/api/Record/Delete/All>
- Request Mode:**
- POST

Authorization

- Bearer token character string

Return example

```
//Returned when verification succeeds

{
  "result": true
}
```

API-Clear records of the specified type

Brief Description:

- Clearing all records is invoked by this interface
- Request URL:**
- <http://Server IP:port/api/Record/Delete/ByType>

Request Mode:

- POST

Authorization

- Bearer token character string

Request required parameter

| Field | Type | Required | Description |
|------------|------|----------|------------------------|
| RecordType | int | Yes | Record Type |
| 1 | | | Identify punch records |
| 2 | | | Door sensor records |
| 3 | | | System records |

Request example

```
{
  "RecordType": 1 //Clear identification punch record
}
```

Return example

```
//Returned when verification succeeds

{
  "result": true
}
```

System file management

API-Get system files

Brief description:

- Call this interface when getting system files
- The system files include: boot image, screensaver image, device camera snapshot

Request URL:

- <http://Server IP:port/api/System/GetFile>

Request method:

- POST

Authorization

- Bearer token character string

Request required parameters

| Field | Type | Length | Required | Description |
|-----------|--------|--------|----------|--|
| FileType | string | | Yes | The type of file that needs to be obtained
BootImage -- boot image
ScreenImage -- screen saver image
Snapshot -- device camera snapshot |
| FileIndex | uint32 | | No | When the request type is ScreenImage,
This parameter needs to be carried, with a
value range of 1-8 |

Request Example

```
{
  "FileType": "ScreenImage",
  "FileIndex": 1
}
```

Return instructions

- If there is a file in the device that needs to be requested, return the binary stream of the file
- If there is no file to be requested in the device, return HTTP response code 404

API-Update system files

Brief description:

- Call this interface when system files need to be replaced
- The system files that can be updated include: boot image, screen saver image
- Update file using form upload method Content-Type: multipart/form-data

Request URL:

- <http://Server IP:port/api/System/UploadFile>

Request method:

- POST

Content-Type

- multipart/form-data

Authorization

- Bearer token character string

Request required parameters

| Field | Type | Length | Required | Description |
|-------|------|--------|----------|-------------|
|-------|------|--------|----------|-------------|

| Field | Type | Length | Required | Description |
|-----------|--------|--------|----------|---|
| FileType | string | | Yes | The type of file that needs to be obtained
BootImage -- BootImage
ScreenImage -- Screen saver image |
| FileIndex | uint32 | | No | When the request type is ScreenImage,,
This parameter needs to be carried, with a
value range of 1-8 |
| FileMD5 | string | | Yes | MD5 of uploaded files |
| File | file | | | Only Jpg images are accepted for device
images
And the size of the image file needs to be
controlled below 1MB |

Request Example

```

POST /api/System/UploadFile HTTP/1.1
Accept: */*
Host: localhost:5000
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Type: multipart/form-data; boundary=-----
-506873351428002157394455
Content-Length: 17842

-----506873351428002157394455
Content-Disposition: form-data; name="FileType"

BootImage
-----506873351428002157394455
Content-Disposition: form-data; name="FileIndex"

1
-----506873351428002157394455
Content-Disposition: form-data; name="FileMD5"

123456abcd123456abcd1263456abcd
-----506873351428002157394455
Content-Disposition: form-data; name="File"; filename="Postman_file.jpg"
Content-Type: image/jpeg

*****jpeg文件二进制内容*****
-----506873351428002157394455--

```

Return Example

```

{
  "result": true,
}

```

API-Remove system file

Brief description:

- Call this interface when deleting system files
- The system files that can be deleted include: screen saver images

Request URL:

- <http://Server IP:port/api/System/DeleteFile>

Request method:

- POST

Authorization

- Bearer token character string

Request required parameters

| Field | Type | Length | Required | Description |
|-----------|--------|--------|----------|---|
| FileType | string | | Yes | The type of file that needs to be obtained
ScreenImage -- Screen saver image |
| FileIndex | uint32 | | No | When the request type is ScreenImage,
This parameter needs to be carried, with a
value range of 1-8 |

Request Example

```
{
  "FileType": "ScreenImage",
  "FileIndex": 1
}
```

*Return Example

```
{
  "result": true,
}
```

API-Obtain the required resolution for screen saver images

Brief description:

- Obtain the required resolution for screensaver files
- The system files that can be deleted include: screensaver images

Request URL:

- <http://Server IP:port/api/System/GetImageSize>

Request method:

- POST

Authorization

- Bearer token character string

Return Example

```
{
  "result": true,
  "content": {
    "width": 720, //Image pixel width
    "Height": 1280 //Pixel height
  }
}
```

##

Return error code

Common error code

| Code Value | Description |
|------------|------------------------|
| :---- | :----- |
| 10000 | Authorization error |
| 10001 | method error |
| 10002 | Content-Type error |
| 10003 | Content-Encoding error |
| 10004 | json parse error |
| 10005 | Content-Length error |